KILNDECK · PROBLEM STATEMENT

KilnDeck problem statement

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	KND-PRB-001	Version	0.4
Status	Draft	Date	2026-10-30
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/01-problem.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-30	Amish Chadha	Piece limit 33 kg (KND-DEC-001 D14, decided 2026-10-30)
0.3	2026-10-28	Amish Chadha	Kit 1.8 users-and-needs and stakeholders tables (Tables 1a and 1b), source of the user requirements KND-URS-001
0.2	2026-10-03	Amish Chadha	TRL 2 and 3 update; kiln geometry stated as working assumptions; open questions answered or carried to the co-design partner (KND-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

KilnDeck problem statement

On most South Asian brick kilns, firing is still a manual job done on foot, on top of the fire. The fuel feeding method has barely changed, and neither has where the fireman stands.

The problem

In a fixed-chimney bull's trench kiln, two firemen standing on top of the kiln feed solid fuel through feed holes every 15 to 20 minutes (Greentech, 2014). They work in pairs, in shifts of six to eight hours, in the sun with no shade, and receive no gloves or masks (The Migration Story).

Existing improvements target fuel, not footing. Stanford's Luby Lab and BUET built motorised automatic coal feeders; pilots found wet coal clogging the feeder and small motors overheating under load (Luby Lab). A 1924 German patent describes a rotating-disc fuel charger for brick kilns, now long expired (US1506924A). None of these give the fireman somewhere safe to stand while he feeds.

Users and context

Table 1. Users.

User	Need	Context
Kiln firemen	Feed fuel at every hole without standing on the crust, and with less heat on the feet	Kiln top in shifts of 6 to 8 hours, day and night, high ambient heat
Kiln owners and managers	A low-cost addition that fits the existing kiln and does not slow production	Seasonal operation, thin margins, local fabrication
Local fabricators	Drawings for track, deck and hopper from stock steel	Small workshops in brick-making districts
Clean-kiln and worker rights programmes	A safety measure that fits alongside fuel-efficiency training	NGO, research and government programmes

Table 1a. Users and needs (kit 1.8; the source of KND-URS-001).

User type	Goals and needs	Typical use scenario	Pain points and challenges
Kiln fireman	Feed every hole in the round on time without stepping on the crust; not fall; less heat on the feet; move and dose fuel by hand	Two firemen in a 6 to 8 h shift, a round every 15 to 20 min, day and night, in summer heat	Weak spots in the crust over the fire, radiant heat, smoke, poor light at night, no gloves or masks
Kiln owner or manager	A cheap addition that fits the existing kiln and does not slow production; the walls must not be damaged	Seasonal firing; the fire moves 6 to 10 m a day round a 180 to 220 m circuit	Thin margins, no power on the kiln top, distrust of equipment that slows feeding
Local fabricator	Drawings and a plan for stock steel, hand tools, a drill and a stick welder; pieces two people can carry	A small workshop in a brick district, fitting the deck on the kiln over the cold zone	Unfamiliar bought parts (wheels, bearings, chains), no lifting gear on site
Clean-kiln or worker rights programme	A safety measure that fits fuel-efficiency training and can be shown to work	Trials with firemen during a firing season	Must not be seen as a reason to work in unsafe heat or storms

Table 1b. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Firemen and their families	The people the deck protects	Interviews on hazards and acceptance; trial users at TRL 4 and later
Kiln owners' associations	Decide whether a kiln fits a deck and pays for it	Survey access to a partner kiln; view on track handling and fuel stock
Clean-kiln research programmes (for example the Stanford and BUET kiln team)	First co-design candidate (KND-DDR-001, D8)	Not yet approached; approach is Amish's decision
Labour inspectorates and environment departments	Kiln licensing and workplace safety rules (KND-REG-001)	To assess; no engagement before TRL 4
Local fabricators and steel stockists	Build and supply the deck	Quotes for the bought parts and steel at TRL 4

Operating environment

- Top of an operating bull's trench kiln: a trench of bricks between an outer wall and an inner wall, covered by an ash crust with feed holes, around a central chimney. The fire moves 6 to 10 m along the kiln a day, on a circuit of 180 to 220 m (Greentech, 2014); whole kilns are about 20 to 30 m wide and 75 to 120 m long (BrickGuru).
- Working assumptions until a survey of the partner kiln (KND-CAL-001, section 1): a trench 6.0 m wide between wall faces; wall tops at least 0.6 m wide and level with the crust within 100 mm; feed holes about 150 mm across, about 1.0 m apart along and across the kiln; crust surface about 150 °C, up to 250 °C at hot spots.
- Radiant and conducted heat from the kiln top; ambient air up to about 45 °C (113 °F) in summer (The Migration Story).
- Coal dust, ash, smoke and rain during the firing season.
- Night work under poor lighting.
- No power at the kiln top.

Constraints

- Value-engineering target for the prototype parts: USD 3,500 (a hypothetical control target, not a spending limit).
- Manual only: no motor; the deck is moved and the fuel is metered by hand.
- Track and supports bear only on sound kiln walls, never on the crust, within a bearing limit set by a survey of each kiln.
- Buildable with hand tools, a drill and a small stick welder from stock steel sections.
- Every piece carried up the kiln weighs 33 kg (73 lb) or less, the HSE two-person guide, so two people can carry it (KND-DEC-001 D14, decided 2026-10-30; the end truss modules are still 34.4 and 34.8 kg, see D16).
- Open design: hardware under CERN-OHL-S-2.0, any calculators under MIT.

> **Safety:** The deck carries a person over a burning kiln. A fall through the guardrail, a deck running off its track, a crushed hand in the drive or the rotor, and heat on the feet are the hazards this design must control. It is published as an open engineering reference, not certified equipment; kiln owners and builders are responsible for their own structural checks and risk assessment.

Out of scope

- Motorised or automatic feeders.
- Kiln conversion (for example to zigzag firing) and emissions control.
- Brick moulding, loading and unloading work.
- Labour rights, wages and bonded labour, which need legal and social action rather than hardware.
- The curved ends of the kiln, in this first prototype (KND-DDR-002, A4).

Prior work

Table 2. Prior work.

Prior work	What it does	Gap for these users	Source
Automatic coal feeder (Luby Lab, Stanford, and BUET)	Motorised funnel feeder for kiln feed holes, piloted in Jessore, Bangladesh	Powered; clogging with wet coal and motor overheating reported; does not give the fireman a safe place to stand	link
US1506924A, fuel-charging device for brick kilns (1924, expired)	Coal hopper with a rotating bottom disc that delivers fuel into the stoking hole	Fixed charger at a single hole; no deck, rails or guardrail	link
Brick kiln intervention trial (Luby Lab)	Training in continuous fuel feeding and better brick setting across 300 kilns	Changes practice, not equipment; firemen still stand on the crust	link
Standard firing practice (Greentech factsheet)	Two firemen on the kiln top feed fuel by hand through feed holes	The status quo that KilnDeck changes	link

Co-design

A clean-kiln programme or kiln owners' association with access to operating kilns and firemen, ideally one already running fuel-feeding training, so the deck can be tried with real firemen during a firing season. First candidate to approach: a clean-kiln research programme in Bangladesh (for example the Stanford and BUET kiln team); second: a kiln owners' association in Punjab, Pakistan (KND-DDR-001, D8). Neither has been approached.

Co-design checklist:

- [ ] Survey of one partner kiln: trench width, wall-top width and condition, crust level and temperature, feed hole grid and lids.
- [ ] Interviews with firemen on the hazards they see and whether they would use a deck.
- [ ] Fuel used at that kiln (coal, biomass or both) and how it reaches the kiln top.
- [ ] Owner's view on where fuel stock is kept and how the track is moved with the fire.

Open questions, answered at TRL 2 and 3

Table 3. The scaffold's open questions.

Question	Answer	Source
Track all the way round, or sections that move with the fire?	Sections on the straight sides, laid ahead of the fire and lifted behind it	KND-DDR-001, D2
How hot do the rails get, and do they need sliding joints?	Yes: slotted fishplates with 6 mm gaps take 3.6 mm per module at +100 K; the deck floats 12 mm on its inner wheels	KND-CAL-001, section 13
Coal and biomass?	Both; about 0.49 kg of coal or 0.15 kg of biomass per rotor pocket (estimates)	KND-CAL-001, section 4
What do firemen see as the hazards, and would they use a deck?	Not a design question; it is the first co-design task above	Co-design
Does the rotor clog with wet coal?	Smooth hopper seams, a 40 mm lump limit and a crank that can be turned back reduce the risk; only a trial can answer it	KND-REQ-001, R5 (TRL 4)